

Sean William Carroll

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Summary

Systems Software Engineer (9 years) who ships performant, concurrent, and memory-conscious C++ and Rust. Builds performance infrastructure, tooling, and compilers.

Skills

Programming: Modern C++, Rust, Python

Tools: Linux, git, CMake, gdb, perf, Valgrind, KCachegrind, clang-tidy, DuckDB, pandas/NumPy

Areas: Profiling, concurrency, serialization, code generation

Experience

AMD Boxborough, MA
SMTS Silicon Design Engineer, Performance Modeling Jul. 2025 – Present

MTS Silicon Design Engineer, Performance Modeling Aug. 2022 – Jul. 2025

- Developed custom binary file format for simulation data aggregations:
 - Wrote a C++ library to efficiently stream data to disk and enforce compile time correctness checking
 - Implemented a Rust application that parsed the format and translated into DuckDB for more efficient analysis of simulation data
- Wrote and owned two large C++ performance models (~10K LOC each), collaborating with other model owners to ensure interoperability of models and identifying common idioms that we lifted into common libraries
- Refactored our main simulation management tool in Rust to use Rust idioms like Result<T, E>, which directly led to the discovery of a race condition causing a child thread to panic and then get silently dropped
- Regularly profiled our C++ simulations to find hotspots and save overall runtime, with one example being the reduction of a 5-hour regression suite to just 2.5 hours of wall clock time fixed by discovering pathological copy constructor and destructor call cycles
- Two papers accepted to AMD's internal technical conference

Redpoint Positioning Corporation Boston, MA
Algorithm Engineer Jan. 2022 – Aug. 2022

- Took prototype Python script from R&D team and transformed it into a library used by multiple internal tools, then optimized the library functions for runtime using NumPy and Pandas, vectorizing many Python-style loops, with one example calculation going from 3 minutes to under 10 seconds
- Developed new Python tool for use by FAEs to more quickly diagnose and troubleshoot malfunctioning deployment sites, saving multiple hours plotting coverage zones manually

Apple Cambridge, MA
Design Verification Engineer, Analog/Mixed Signal Group Aug. 2021 – Jan. 2022

- Wrote assertions to ensure specific invariants were never violated during DV simulations

Lightelligence Boston, MA
Computer Architect, Performance Modeling Mar. 2020 – Aug. 2021

- Owned greenfield functional and performance simulator for AI inference chip written in C++, used by both the runtime development team and by the hardware team as a golden reference model
- Optimized proprietary compiler written in a mix of Python and C++ to recognize repeating execution graph structures and replace them with proprietary instructions, effectively utilizing our custom coprocessor to accelerate matrix multiplies

Design Verification Engineer Nov. 2019 – Mar. 2020

- Developed a code generation toolchain in Python to generate collateral for RTL (Verilog), DV (SystemVerilog), and Embedded (C/C++) teams from a single source of truth, leveraging an existing SystemRDL parser, with a later extension allowing reuse by technical writers in documentation generation

Marvell Semiconductor (formerly Cavium) Marlborough, MA
Design Verification Engineer Sep. 2017 – Nov. 2019

- Coauthored lightweight OS framework in C for testing basic customer use cases during pre-silicon development, bringing up a lightweight scheduler before running driver code to configure and interact with coprocessor subsystems

Education

Georgia Institute of Technology Atlanta, GA
M.S. in Computer Science Jan. 2018 – May 2021

- OMSCS (Remote), GPA: 4.0, Machine Learning Specialization

Cornell University Ithaca, NY
B.S. in Computer Science, Electrical and Computer Engineering Aug. 2013 – May 2017

- GPA: 3.871 (Magna Cum Laude), Dean's List (All Semesters), Systems/Databases Vector (OS Track)